Topic: Bernoulli equations

Question: Find the solution to the Bernoulli differential equation.

$$y' + \frac{y}{x} = xy^2$$

Answer choices:

A $y = \dfrac{1}{-x^2 + Cx}$

B $y = -x^2 + Cx$

C $y = x^2 + C$

D $y = \dfrac{1}{\sqrt{-x^2 + Cx}}$

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Solution: A

With the equation in standard form, divide through by y^n. In this equation, that means we're dividing by y^2.

$$y' + \frac{y}{x} = xy^2$$

$$\frac{y'}{y^2} + \frac{y}{xy^2} = \frac{xy^2}{y^2}$$

$$y^{-2}y' + \frac{1}{x}y^{-1} = x$$

Our substitution is $v = y^{-1}$, so we'll differentiate to get

$$v' = -y^{-2}y'$$

and then solve this for $y^{-2}y'$.

$$y^{-2}y' = -v'$$

Now we can make substitutions into the Bernoulli equation.

$$y^{-2}y' + \frac{1}{x}y^{-1} = x$$

$$-v' + \frac{1}{x}v = x$$

Multiplying through by -1 puts the equation in standard form of a linear differential equation.

$$v' - \frac{1}{x}v = -x$$

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To solve the linear equation, we'll use $P(x) = -1/x$ to find the integrating factor,

$$I(x) = e^{\int P(x)\,dx}$$

$$I(x) = e^{\int -\frac{1}{x}\,dx}$$

$$I(x) = e^{-\ln x}$$

$$I(x) = e^{\ln(x^{-1})}$$

$$I(x) = x^{-1}$$

and then multiply through the linear equation by $I(x)$.

$$v' - \frac{1}{x}v = -x$$

$$x^{-1}v' - \frac{1}{x}vx^{-1} = -x^{-1}x$$

$$\frac{1}{x}v' - \frac{1}{x^2}v = -1$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(vx^{-1}) = -1$$

then integrate both sides.

$$\int \frac{d}{dx}(vx^{-1}) = \int -1\,dx$$

$$vx^{-1} = -x + C$$

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Solve for v.

$$v = -x^2 + Cx$$

Use $v = y^{-1}$ to back-substitute for v,

$$y^{-1} = -x^2 + Cx$$

then solve for y.

$$\frac{1}{y} = -x^2 + Cx$$

$$y = \frac{1}{-x^2 + Cx}$$

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Topic: Bernoulli equations

Question: If $y(1) = 1$, find the solution to the Bernoulli differential equation.

$$xy' - y = y^3 \ln x$$

Answer choices:

A $y = \dfrac{1}{\ln x + 1}$

B $y = \dfrac{1}{\sqrt{-\ln x + 1}}$

C $y = \dfrac{1}{2} - \ln x + \dfrac{1}{2}x^{-2}$

D $y = \dfrac{1}{\sqrt{\frac{1}{2} - \ln x + \frac{1}{2}x^{-2}}}$

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Solution: D

Start by rewriting the equation in standard form.

$$xy' - y = y^3 \ln x$$

$$y' - \frac{y}{x} = \frac{y^3 \ln x}{x}$$

With the equation in standard form, divide through by y^n. In this equation, that means we're dividing by y^3.

$$\frac{y'}{y^3} - \frac{y}{xy^3} = \frac{y^3 \ln x}{xy^3}$$

$$y'y^{-3} - \frac{1}{x}y^{-2} = \frac{\ln x}{x}$$

Our substitution is $v = y^{-2}$, so we'll differentiate to get

$$v' = -2y^{-3}y'$$

and then solve this for $y^{-3}y'$.

$$y^{-3}y' = -\frac{1}{2}v'$$

Now we can make substitutions into the Bernoulli equation.

$$y'y^{-3} - \frac{1}{x}y^{-2} = \frac{\ln x}{x}$$

$$-\frac{1}{2}v' - \frac{1}{x}v = \frac{\ln x}{x}$$

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Multiplying through by -2 puts the equation in standard form of a linear differential equation.

$$v' + \frac{2}{x}v = -\frac{2\ln x}{x}$$

To solve the linear equation, we'll use $P(x) = 2/x$ to find the integrating factor,

$$I(x) = e^{\int P(x)\,dx}$$

$$I(x) = e^{\int \frac{2}{x}\,dx}$$

$$I(x) = e^{2\ln x}$$

$$I(x) = e^{\ln x^2}$$

$$I(x) = x^2$$

and then multiply through the linear equation by $I(x)$.

$$v' + \frac{2}{x}v = -\frac{2\ln x}{x}$$

$$x^2v' + \frac{2}{x}x^2v = -\frac{2\ln x}{x}x^2$$

$$x^2v' + 2xv = -2x\ln x$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(vx^2) = -2x\ln x$$

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then integrate both sides. We'll use integration by parts on the right with $u = \ln x$, $du = (1/x)\ dx$, $dv = -2x\ dx$, and $v = -x^2$.

$$\int \frac{d}{dx}(vx^2) = \int -2x \ln x\ dx$$

$$vx^2 = -x^2 \ln x - \int -x^2 \left(\frac{1}{x}\right)\ dx$$

$$vx^2 = -x^2 \ln x + \int x\ dx$$

$$vx^2 = \frac{1}{2}x^2 - x^2 \ln x + C$$

Solve for v.

$$v = \frac{1}{2} - \ln x + Cx^{-2}$$

Use $v = y^{-2}$ to back-substitute for v,

$$y^{-2} = \frac{1}{2} - \ln x + Cx^{-2}$$

then solve for y.

$$\frac{1}{y^2} = \frac{1}{2} - \ln x + Cx^{-2}$$

$$y^2 = \frac{1}{\frac{1}{2} - \ln x + Cx^{-2}}$$

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$$y = \frac{1}{\sqrt{\frac{1}{2} - \ln x + Cx^{-2}}}$$

Once we have this general solution, we recognize from the initial condition $y(1) = 1$ that $x = 1$ and $y = 1$, so we'll plug these values into the general solution to find C.

$$1 = \frac{1}{\sqrt{\frac{1}{2} - \ln 1 + 1^{-2}C}}$$

$$1 = \frac{1}{\sqrt{\frac{1}{2} + C}}$$

$$1 = \frac{1}{2} + C$$

$$C = \frac{1}{2}$$

Then the solution is

$$y = \frac{1}{\sqrt{\frac{1}{2} - \ln x + \frac{1}{2}x^{-2}}}$$

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Topic: Bernoulli equations

Question: If $y(0) = 1$, find the solution to the Bernoulli differential equation.

$$y' + y = x^2 y^2$$

Answer choices:

A $$y = \frac{1}{x^2 + 2x + 2 - e^{-x}}$$

B $$y = \frac{1}{x^2 + 2x + 2 - e^{x}}$$

C $$y = x^2 + 2x + 2 - e^{x}$$

D $$y = \frac{1}{x^2 + 2x + 2}$$

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Solution: B

With the equation in standard form, divide through by y^n. In this equation, that means we're dividing by y^2.

$$y' + y = x^2y^2$$

$$\frac{y'}{y^2} + \frac{y}{y^2} = \frac{x^2y^2}{y^2}$$

$$y^{-2}y' + y^{-1} = x^2$$

Our substitution is $v = y^{-1}$, so we'll differentiate to get

$$v' = -y^{-2}y'$$

and then solve this for $y^{-2}y'$.

$$y^{-2}y' = -v'$$

Now we can make substitutions into the Bernoulli equation.

$$y^{-2}y' + y^{-1} = x^2$$

$$-v' + v = x^2$$

Multiplying through by -1 puts the equation in standard form of a linear differential equation.

$$v' - v = -x^2$$

To solve the linear equation, we'll use $P(x) = -1$ to find the integrating factor,

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$$I(x) = e^{\int P(x)\,dx}$$

$$I(x) = e^{\int -1\,dx}$$

$$I(x) = e^{-x}$$

and then multiply through the linear equation by $I(x)$.

$$e^{-x}y' - e^{-x}y = -e^{-x}x^2$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(ye^{-x}) = -x^2e^{-x}$$

then integrate both sides. We'll use integration by parts on the right with $u = -x^2$, $du = -2x\,dx$, $dv = e^{-x}\,dx$, and $v = -e^{-x}$.

$$\int \frac{d}{dx}(ye^{-x})\,dx = \int -x^2e^{-x}\,dx$$

$$ye^{-x} = x^2e^{-x} - 2\int xe^{-x}\,dx$$

We'll apply integration by parts again on the right with $u = x$, $du = dx$, $dv = e^{-x}\,dx$, and $v = -e^{-x}$.

$$ye^{-x} = x^2e^{-x} - 2\left(-xe^{-x} + \int e^{-x}\,dx\right)$$

$$ye^{-x} = x^2e^{-x} + 2xe^{-x} + 2e^{-x} + C$$

Solve for y.

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$$v = x^2 + 2x + 2 + Ce^x$$

Use $v = y^{-1}$ to back-substitute for v,

$$y^{-1} = x^2 + 2x + 2 + Ce^x$$

then solve for y.

$$\frac{1}{y} = x^2 + 2x + 2 + Ce^x$$

$$y = \frac{1}{x^2 + 2x + 2 + Ce^x}$$

Once we have this general solution, we recognize from the initial condition $y(0) = 1$ that $x = 0$ and $y = 1$, so we'll plug these values into the general solution to find C.

$$1 = \frac{1}{0^2 + 2(0) + 2 + Ce^0}$$

$$1 = \frac{1}{2 + C}$$

$$1 = 2 + C$$

$$C = -1$$

Then the solution is

$$y = \frac{1}{x^2 + 2x + 2 - e^x}$$

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